

Rates of Change and Behavior of Graphs

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: -2

_____ Find the average rate of change of $f(x) = x^2$ on the interval $[1, 3]$.

ANSWER: 3

_____ For the graph above, state the local minimum value.

ANSWER: 4

_____ Find the average rate of change of $f(x) = x^2$ on the interval $[2, 5]$.

ANSWER: 7

_____ Find the average rate of change of $f(x) = x^2 - \frac{1}{x}$ on the interval $[2, 4]$.

ANSWER: 5 # _____ Find the average rate of change of $f(x) = x^3$ on the interval $[1, 3]$.	ANSWER: $\frac{49}{8}$ # _____ Find the average rate of change of $f(x) = 5x - 1$ on the interval $[0, 10]$.
ANSWER: 1 # _____ For the graph above, state the local maximum value.	ANSWER: $(1, 4)$ # _____ For the same graph, on which interval(s) is it increasing? (interval notation)
ANSWER: $(-\infty, 1) \cup (4, \infty)$ # _____ Find the average rate of change of $f(x) = x^2 + 2$ on the interval $[-1, 2]$.	ANSWER: 13 # _____ A graph rises to a peak at $(1, 3)$, falls to a valley at $(4, -2)$, then rises again. On which interval is it decreasing? (interval notation)